PRODUCT REQUIREMENTS DOCUMENT

QueryTrust AI: Governed Natural-Language Analytics Workbench

Trustworthy NL-to-SQL, Multi-Layer AST Safety, Telemetry Outlier Detection, and Regression Benchmark Suite

Independent student capstone project | Version 1.0 | 14 October 2026

1. Problem

Enterprise business users and operations engineers frequently need answers from relational databases (e.g., asset counts, warranty schedules, incident trends, regional hardware health) but lack SQL proficiency, relational schema knowledge, or statistical tools. While modern Large Language Models (LLMs) can translate natural language into SQL, naive text-to-SQL implementations introduce catastrophic risks:

1. **Security Vulnerabilities:** Prompt-based instructions fail as security boundaries. Malicious or hallucinated queries can execute destructive DDL/DML (`DROP` , `DELETE` , `UPDATE`), access sensitive system catalogs (`pg_catalog` , `information_schema`), perform out-of-band data exfiltration (`dblink` , `pg_read_file`), or trigger Denial-of-Service (`pg_sleep` , unbounded cross-joins).
2. **Semantic Hallucinations & Silent Logic Errors:** LLMs invent nonexistent tables/columns, misinterpret undefined business concepts (e.g., fabricating arbitrary logic for "high-risk" or "unhealthy"), and assume financial metrics (revenue, profit, churn) exist within purely operational hardware schemas.
3. **Execution Opacity & Runaway Queries:** Standard chat interfaces execute generated queries immediately and blindly, without letting users preview the SQL, inspect AST validation findings, or understand assumptions. Runaway queries without row limits or execution timeouts lock database connections.
4. **Lack of Verifiable Measurement:** Teams deploy generative analytics without regression test suites or baseline-versus-guarded comparisons, relying on cherry-picked demos and uncalibrated claims.

Project Scope & Academic Boundary

QueryTrust AI builds a local-first, full-stack, governed natural-language analytics workbench. It demonstrates that combining schema-aware semantic context, formal AST parsing via SQLGlot, multi-layer policy enforcement, dual database roles with strict `SELECT`-only privileges, and a repeatable regression benchmark delivers trustworthy analytics. This student project runs self-contained without proprietary corporate infrastructure, customer datasets, or paid cloud services. Real Cisco or client data, logos, or commercial affiliations are strictly excluded; all data is deterministically synthesized for educational evaluation.

2. Product Goal

Build a production-grade, local full-stack web application where an operator, analyst, or administrator can:

1. **Explore Hardware Fleet Intelligence:** Inspect KPI summaries, regional distributions, warranty expiration horizons, and open service incident trends over synthetic networking assets.
2. **Ask Natural-Language Data Questions:** Formulate questions in plain English (e.g., "Show active routers in the North region with warranties expiring next month").
3. **Preview & Inspect Governed SQL Before Execution:** Review the candidate SQL query, interpreted intent, assumptions, and a multi-layer Abstract Syntax Tree (AST) safety checklist before any database interaction occurs.
4. **Experience Deterministic Disambiguation:** Receive structured clarification options when requesting undefined business terms (e.g., "high-risk devices") and clear out-of-domain rejections for unsupported concepts (e.g., revenue, churn) without hallucinated SQL.
5. **Execute Under Dual-Layer Sandboxing:** Run server-stored, validated query previews via a dedicated PostgreSQL read-only database role (`analytics_reader`) with strict row clamping and statement timeouts.
6. **Profile Data Quality:** Run deterministic data profiling across asset tables to detect missing fields, invalid date sequences, inconsistent categories, duplicate serial numbers, and telemetry range violations.
7. **Monitor Telemetry Outliers:** Scan device telemetry using a genuine scikit-learn `IsolationForest` model to flag multivariate operational anomalies, explicitly presented as statistical outliers rather than device failure probabilities.

8. **Evaluate & Benchmark Reliability:** Execute a 30+ case automated evaluation suite comparing a raw LLM baseline against the guarded QueryTrust pipeline, measuring execution correctness, safe rejection rate, clarification accuracy, and latency.

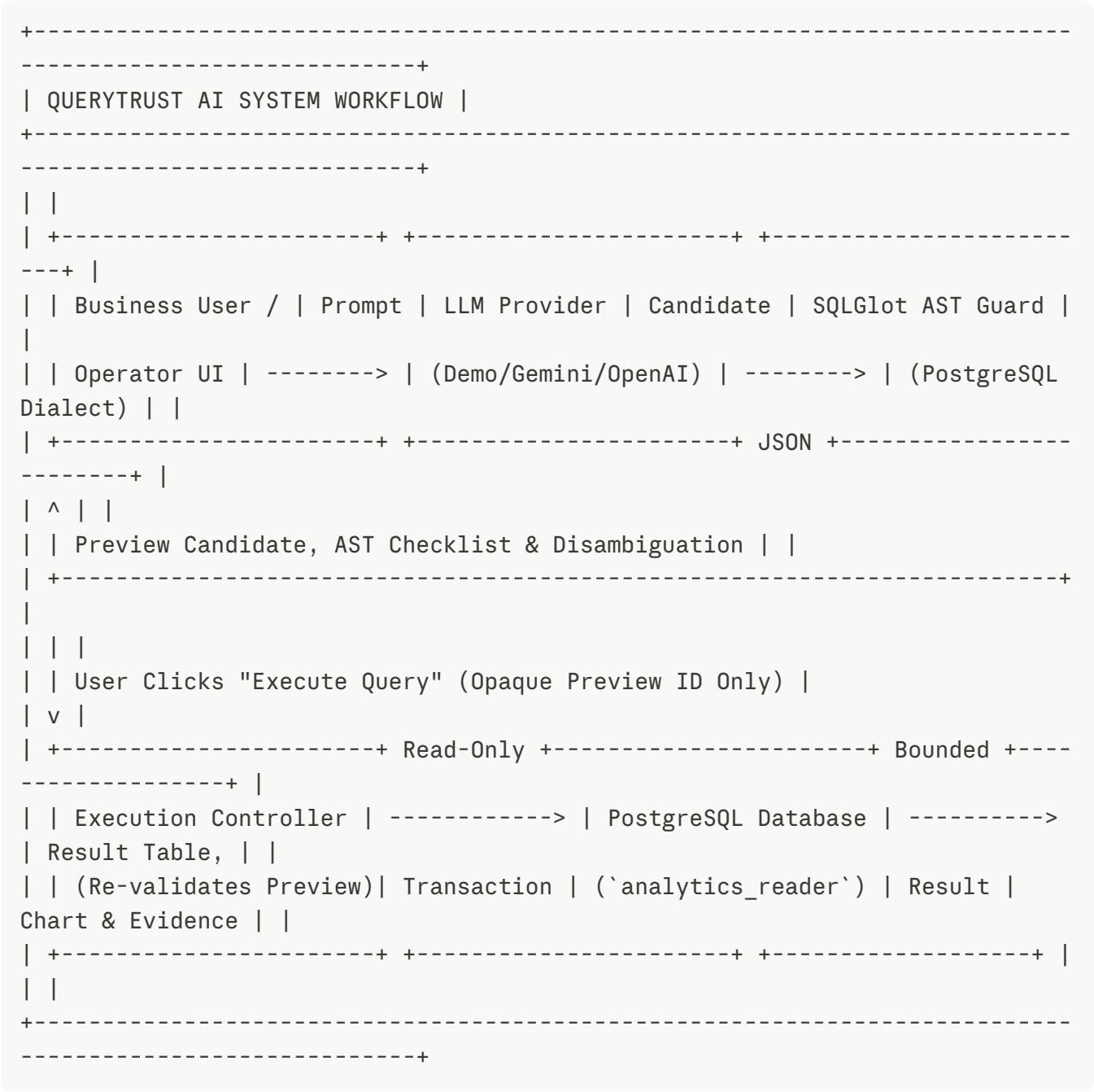


Figure 1. The natural-language query path enforces complete separation between preview translation and governed execution.

System Invariants & Truthfulness Rules

- **Natural-Language Path is Read-Only for All Roles:** Even System Administrators cannot execute data-modifying SQL through natural language.
- **Never Fabricate Execution or Model Metrics:** If no live LLM key is configured, the application operates in an honest, visibly badged **Demo Mode**. Accuracy metrics, anomaly scores, and test passes are never hard-coded or faked.
- **Preview ID Execution Only:** The execution API accepts only an opaque, server-stored preview ID (`query_run_id`). The browser cannot submit raw SQL for execution.
- **IsolationForest is NOT Failure Prediction:** Telemetry anomalies are labeled strictly as statistical multivariate outliers; they are never represented as probabilities of hardware failure.

3. Users and Core Stories

Role	Core Purpose	Access Boundary
Viewer	Operational staff reviewing hardware inventory and asking read-only questions	Reads dashboard, devices, sites; asks NL queries; views own history. Cannot run profiling, anomaly scans, admin tasks, or write to database.
Analyst	Technical lead diagnosing fleet health, data hygiene, and telemetry	All Viewer capabilities, plus initiates Data Quality profiling, runs Telemetry Anomaly scans, and submits query feedback.
Administrator	System owner auditing security, managing users, and verifying reliability	All Analyst capabilities, plus user management, security audit log access, and execution of the Evaluation Lab regression benchmark.
Assessor / Reviewer	Academic or technical evaluator validating project requirements	Evaluates clean-clone setup, automated test suite, security injection blocks, demo mode, and live execution.

Core User Stories

1. As a Viewer, I want to ask questions about our network devices in plain English so that I can find operational answers without knowing SQL syntax.
2. As a Viewer, I want to see the interpreted intent and generated SQL before execution so that I can verify the system understood my question correctly.
3. As an Analyst, I want the system to ask for clarification when I use ambiguous terms like "high-risk" so that business decisions are not based on hidden assumptions.
4. As an Analyst, I want the system to decline questions about revenue or churn immediately so that I know the boundary of the operational inventory dataset.
5. As an Analyst, I want to profile the device database for missing warranties, bad dates, and duplicate serials so that dirty data can be identified and remediated.
6. As an Analyst, I want to run an anomaly scan over telemetry readings so that devices exhibiting unusual temperature, latency, or packet loss can be investigated.
7. As an Administrator, I want all natural-language queries executed through a restricted, `SELECT`-only database role so that our core tables can never be altered or dropped.
8. As an Administrator, I want to run an automated regression benchmark so that I can measure whether prompt or code changes improve accuracy or introduce regressions.
9. As an Assessor, I want to start the application with a single Docker Compose command and run tests in offline demo mode without needing paid API keys.

4. Required Implementation Stack

Layer	Required Choice	Purpose / Architectural Rationale
Frontend Framework	React 18/19, TypeScript, Vite	Fast, typed, reactive single-page application matching modern web standards.
Frontend Styling	Custom CSS Design System / Modern Vanilla CSS tokens	Ultra-crisp, responsive, enterprise-grade dark/light theme without bulky framework bloat.
Icons & Charts	Lucide React, SVG visualization	Scalable, clean iconography and responsive tabular/chart data rendering.

Layer	Required Choice	Purpose / Architectural Rationale
Backend Framework	Python 3.12 / 3.13, FastAPI, Uvicorn	Async, high-performance REST API with automatic OpenAPI documentation and strict typing.
Data Validation	Pydantic v2	Robust, type-safe serialization, deserialization, and schema enforcement for all endpoints.
Database	PostgreSQL 16+	Production-grade relational database supporting foreign keys, transactional integrity, and granular RBAC.
ORM & Migrations	SQLAlchemy 2.x, Alembic	Declarative database models, connection pooling, and version-controlled schema migrations.
Database Security	Dual PostgreSQL Roles (`app_user`, `analytics_reader`)	Hardened defense-in-depth: `analytics_reader` possesses zero `INSERT` / `UPDATE` / `DELETE` / DDL privileges.
SQL AST Guard	SQLGlot (PostgreSQL dialect)	Deep AST inspection, single-statement verification, clause inspection, table allowlisting, and row limit clamping.
ML Inference	scikit-learn `IsolationForest`, NumPy, Pandas	Unsupervised outlier detection on device telemetry, with serialized model artifacts and metadata tracking.
Authentication	Passlib (Bcrypt), Python-JOSE (JWT)	Secure password hashing, signed short-lived access tokens, and backend dependency-based RBAC.
Testing	pytest, pytest-asyncio, HTTPX	Comprehensive automated unit, integration, and security test suites with zero external network dependencies.

Layer	Required Choice	Purpose / Architectural Rationale
Containerization	Docker, Docker Compose	Reproducible multi-service deployment (PostgreSQL, backend, frontend) with volume persistence and health checks.

Local Prerequisites

- **Python:** 3.12+ or 3.13+ (tested with 3.13.9)
- **Node.js & npm:** Node.js 20+ (tested with v24.14.1), npm 10+ (tested with 11.11.0)
- **Database:** Local PostgreSQL 16+ or Docker Compose
- **Git:** Standard git client

Required Repository Structure

```
querytrust-ai/
├── docker-compose.yml # Multi-container local orchestration (Postgres,
Backend, Frontend)
├── .env.example # Documented template for all required runtime environment
variables
├── .gitignore # Excludes .env, pycache, dist, and local artifacts
├── AGENTS.md # Persistent engineering invariants and rules
├── PROJECT_SPEC.md # Master project specification
├── ANTIGRAVITY_MASTER_PROMPT.md # Multi-milestone execution plan
├── README.md # Comprehensive setup, run, and evaluation documentation
├── docs/
| ├── PRD.md # This official Product Requirements Document
| ├── BUILD_STATUS.md # Verifiable milestone completion record and test logs
| ├── ARCHITECTURE.md # Detailed component diagrams and data flow contracts
| ├── SECURITY.md # Threat model, AST validation rules, and RBAC matrix
| └── EVALUATION_PLAN.md # Benchmark test cases, metrics, and regression
methodology
├── backend/
| ├── pyproject.toml # Python project metadata, dependencies, and pytest
configuration
| ├── alembic.ini # Alembic migration configuration
| ├── alembic/ # Schema migration scripts
| ├── app/
| | ├── main.py # FastAPI application factory, middleware, and router
registration
```

```
| | ├── config.py # Typed environment settings (Pydantic BaseSettings)
| | ├── database.py # Dual engine setup: app_user (read/write) &
analytics_reader (read-only)
| | ├── core/ # Security, settings, logging, SQL validator, semantic catalog,
LLM provider
| | | ├── auth.py # Password hashing, JWT token creation, and auth
dependencies
| | | ├── sql_validator.py # SQLGlot AST safety validator and row-clamp
engine
| | | ├── semantic_catalog.py # Business metrics, table definitions, approved
joins, ambiguity rules
| | | ├── llm_provider.py # BaseLLMProvider, DemoLLMProvider,
GeminiLLMProvider, OpenAICompatible
| | | ├── data_quality.py # 6-rule deterministic data profiling engine
| | | └── anomaly_detector.py # IsolationForest fitting, joblib
serialization, and telemetry scoring
| | ├── models/ # SQLAlchemy declarative models
| | | ├── user.py # User and RBAC model
| | | ├── device.py # Device, Site, ServiceEvent models
| | | ├── telemetry.py # TelemetryReading model
| | | ├── chat.py # ChatSession, ChatMessage models
| | | ├── query.py # QueryRun, QueryFeedback models
| | | ├── data_quality.py # DataQualityRun, DataQualityFinding models
| | | ├── anomaly.py # MLModelVersion model
| | | └── evaluation.py # EvaluationRun model
| | ├── schemas/ # Pydantic v2 request and response schemas
| | | ├── auth.py # Login, Token, UserResponse schemas
| | | ├── analytics.py # PreviewRequest, PreviewResponse, ExecuteRequest,
ExecuteResponse
| | | ├── chat.py # ChatSessionCreate, ChatMessageResponse schemas
| | | ├── devices.py # DeviceSummary, DeviceDetail, SiteSummary schemas
| | | ├── data_quality.py # DataQualityRunResponse, Finding schemas
| | | ├── anomalies.py # TelemetryScanRequest, AnomalyResult, ModelMetadata
schemas
| | | └── evaluations.py # EvaluationRunRequest, EvaluationRunResponse
schemas
| | ├── evaluation/ # Evaluation Lab & Regression Benchmark
| | | ├── benchmark_suite_v1.json # 37 versioned test cases across 9
categories
| | | ├── benchmark_runner.py # Deterministic offline & live benchmark runner
| | | └── run_cli.py # CLI runner entrypoint
| | ├── api/v1/ # Versioned REST API endpoints
| | | ├── auth.py # /api/v1/auth (login, me)
```

```
| | | ├── dashboard.py # /api/v1/dashboard (summary KPI cards and
distributions)
| | | ├── devices.py # /api/v1/devices (paginated catalog, search, filters)
| | | ├── sites.py # /api/v1/sites (site list and details)
| | | ├── analytics.py # /api/v1/analytics (preview, execute, feedback,
history)
| | | ├── chat.py # /api/v1/chat/sessions (conversation management and
message retrieval)
| | | ├── data_quality.py # /api/v1/data-quality (run, summary, findings)
| | | ├── anomalies.py # /api/v1/anomalies (scan, summary, train)
| | | ├── evaluations.py # /api/v1/evaluations (run, results, history)
| | | └── admin.py # /api/v1/admin (users, audit logs)
| | └── seed/ # Deterministic synthetic data generation
| | └── seed_data.py # Generates 5,000 devices, 50 sites, 6,000 service
events, 25k telemetry
| └── tests/ # Automated test suite (95 tests across 15 files)
| ├── conftest.py # Test database fixtures and mock clients
| ├── test_auth.py # 13 tests: Authentication, JWT, and RBAC
| ├── test_dashboard.py # 2 tests: Fleet KPI summary and distributions
| ├── test_devices.py # 9 tests: Device pagination, search, filters
| ├── test_health.py # 4 tests: Health and readiness probes
| ├── test_sql_validator.py # 16 tests: SQLGlot AST validation and security
injections
| ├── test_llm_provider.py # 7 tests: LLM provider and demo rule engine
| ├── test_analytics_preview.py # 6 tests: Preview generation and
clarification
| ├── test_analytics_execute.py # 6 tests: Read-only execution and preview ID
enforcement
| ├── test_analytics_feedback.py# 3 tests: Query feedback submission
| ├── test_analytics_history.py # 3 tests: User query audit history
| ├── test_chat_sessions.py # 4 tests: Session ownership isolation
| ├── test_data_quality.py # 6 tests: 6-rule data profiling verification
| ├── test_anomaly_detector.py# 6 tests: IsolationForest inference and
metadata
| ├── test_evaluation.py # 9 tests: 37-case benchmark runner and API
| └── test_database_security.py# 1 test: Database kernel SELECT-only
enforcement
├── frontend/
| ├── package.json # React, Vite, TypeScript dependencies
| ├── tsconfig.json # Strict TypeScript compiler options
| ├── vite.config.ts # Vite dev server and proxy configuration
| ├── nginx.conf # Production reverse-proxy and SPA routing config
| ├── Dockerfile # Multi-stage production container build
```

```
| ├── index.html # HTML entry point with modern typography
| └── src/
| ├── main.tsx # React DOM mount point
| ├── App.tsx # Root router, global layout, navigation shell, user context
| ├── index.css # Comprehensive design system tokens, typography, and
component styling
| ├── api/
| | └── client.ts # Strictly-typed HTTP client wrapping all /api/v1 endpoints
| └── components/
| ├── OverviewView.tsx # KPI summary cards, warranty horizons, incident
distributions
| ├── DevicesView.tsx # Paginated device list, status filters, search, modal
details
| ├── AskDataView.tsx # Natural-language workbench, AST checklist, SQL
preview, execution
| ├── QueryHistoryView.tsx# User query audit trail, execution durations,
feedback
| ├── DataQualityView.tsx # Data hygiene findings table, severity filters,
sample inspector
| ├── AnomalyMonitorView.tsx # Telemetry anomaly scatter, outlier ranking,
model metadata
| ├── EvaluationLabView.tsx # Benchmark test runner, regression comparator,
pass-rate metrics
| ├── EvaluationLabView.css # Dedicated styles for Evaluation Lab component
| └── AdminView.tsx # User management, role transitions, security audit logs
└── scripts/
├── setup_db.sql # PostgreSQL database and dual-role setup
├── seed.ps1 / seed.sh # Database migration and seeding helpers
├── run_evaluation.ps1 / .sh # CLI regression benchmark runners
└── generate_prd_pdf.py # PRD to PDF compiler with headless Chrome/Edge
```

5. AI Access, Semantic Catalog, and Local Mode

5.1 Pluggable LLM Provider Architecture

The application defines a strict `BaseLLMProvider` interface to decouple business logic from external model vendors:

1. **`DemoLLMProvider` (Default / Local Mode):**
   - Operates 100% offline without external network calls or API credentials.

    - Powered by a deterministic semantic parser covering fleet aggregations, regional filtering, warranty horizons, incident triage, ambiguity disambiguation, out-of-domain detection, and injection testing.
    - The UI displays an honest badge: `Mode: Demo (Deterministic Rule Engine)`. It never claims a live LLM was invoked.
2. **`GeminiLLMProvider`:**
    - Integrates Google Gemini API using `GEMINI_API_KEY`.
    - Uses structured outputs (`response_mime_type="application/json"`) with single-turn repair retries on malformed syntax.
3. **`OpenAICompatibleLLMProvider`:**
    - Connects to OpenAI or any compatible local endpoint (e.g., vLLM, Ollama, Groq) via `OPENAI_API_KEY` and `OPENAI_BASE_URL`.

Security Rule: API keys are stored exclusively in backend environment variables (`.env`). Keys are never sent to the browser, logged in application outputs, or written into audit artifacts.

5.2 Central Semantic Catalog

Located in `app/core/semantic_catalog.py`, the semantic catalog serves as the single source of truth for database schema definitions, joins, and business terms:

- **Allowed Tables:** Strictly `devices`, `sites`, `service_events`.
- **Approved Joins:**
    - `devices.site_id > sites.id`
    - `service_events.device_id > devices.id`
- **Formal Business Metric Definitions:**
    - Device Count: `COUNT(DISTINCT devices.id)`
    - Active Device: `LOWER(TRIM(devices.status)) = 'active'`
    - Open Service Event: `LOWER(TRIM(service_events.status)) NOT IN ('resolved', 'closed', 'cancelled')`
    - Warranty Expiring in N Days: `devices.warranty_expiry >= '2026-10-14' AND devices.warranty_expiry <= ('2026-10-14'::date + INTERVAL 'N days')` (uses fixed reference date for reproducible evaluation).
- **Ambiguity Disambiguation Rules:**

    - The term "high-risk" has no intrinsic definition. The catalog mandates returning `intent_status: clarification_needed` with explicit choices:
        1. Imminent Warranty Expiry (`< 30 days`)
        2. Open Critical Service Incidents (`severity = 'critical'`)
        3. Offline / Degraded Status
- **Explicit Unsupported Domain Rejections:**
    - Concepts such as `revenue`, `profit`, `cost`, `churn`, and `salary` are defined as unsupported. The catalog mandates returning `intent_status: unsupported` with an explanation that the database is an operational networking inventory, preventing hallucinations.

6. Student Workflow and Screens

The workbench follows an enterprise-grade operator workflow across 9 functional screens:

Step	Screen	Required Behavior
1	Authentication & Shell	Sign in with email/password; development mode provides quick-switch demo pills (`Admin`, `Analyst`, `Viewer`). Top banner displays current user role, live LLM mode, and synthetic data disclosure.
2	Overview Dashboard	Displays 5 live KPI cards (Total Devices, Active Fleet, 30-Day Expiring Warranties, Open Incidents, Quality Issues). Renders device status distribution, warranty expiry trend, and open incidents by severity from live PostgreSQL data.
3	Device Catalog	Paginated inventory table with search, device type filters (`switch`, `router`, `access_point`, `firewall`), status filters, and site location mappings.

Step	Screen	Required Behavior
4	Ask Data (Workbench)	Multi-turn chat interface with left conversation session drawer. Starter prompt chips for fast testing. Displays interpreted intent, clarification buttons, unsupported domain notices, SQL preview, AST checklist, and "Execute Candidate Query" action.
5	Execution & Visuals	Renders query results in a responsive data grid with row count and duration badges. Automatically recommends and renders charts (bar, pie, metric card) for aggregate queries. Provides "Helpful / Not Helpful" feedback widget.
6	Data Quality	Runs deterministic quality profiling across assets. Displays summary hygiene score, categorized findings table (Missing Fields, Invalid Dates, Category Inconsistencies, Duplicate Serials), severity badges, and remediation suggestions.
7	Anomaly Monitor	Runs IsolationForest telemetry scan over 30,000+ data points. Renders device telemetry scatter plot, top-anomalies data grid (CPU, Memory, Temp, Packet Loss, Latency), model version, and synthetic outlier disclaimer.
8	Evaluation Lab (Admin)	Admin-only suite running 30+ benchmark cases. Displays execution correctness %, safe rejection rate %, multi-turn accuracy %, latency percentiles, and test-by-test failure diagnostics.
9	Administration (Admin)	Admin-only user management table, active status toggles, role transitions, and immutable security audit log viewer.

Visual Direction and UI Quality Bar

- **Design System:** Deep slate/navy shell (`#0f172a`, `#1e293b`), crisp neutral white/gray workspace surfaces, restrained indigo (`#6366f1`) and teal (`#14b8a6`) accents, and emerald/amber/rose semantic status badges.

- **Typography:** Modern clean sans-serif (Inter / System UI) with strict hierarchical sizing, monospace code blocks (`Fira Code` , `JetBrains Mono`), and high-contrast readability.
- **Responsiveness & Accessibility:** Fully responsive design tested at 390px (mobile), 768px (tablet), and 1440px (desktop). Keyboard navigation support (Enter to submit query, Shift+Enter for newline, accessible modal dialogs, visible focus outlines).
- **State Handling:** Real skeleton loaders, non-empty initial states with starter chips, actionable error banners with diagnostic details, and zero silent failures or placeholder buttons.

7. Core Data Model and Synthetic Dataset Contract

7.1 Database Schema Specifications

1. Users & RBAC

```
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email VARCHAR(255) UNIQUE NOT NULL,
    display_name VARCHAR(100) NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(20) NOT NULL CHECK (role IN ('admin', 'analyst', 'viewer')),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

2. Sites / Locations

```
CREATE TABLE sites (
    id SERIAL PRIMARY KEY,
    site_code VARCHAR(20) UNIQUE NOT NULL,
    site_name VARCHAR(100) NOT NULL,
    city VARCHAR(100) NOT NULL,
    region VARCHAR(50) NOT NULL
);
```

3. Devices (Hardware Inventory)

```
CREATE TABLE devices (
    id SERIAL PRIMARY KEY,
    asset_tag VARCHAR(50) UNIQUE NOT NULL,
    device_type VARCHAR(50) NOT NULL,
    vendor VARCHAR(50) NOT NULL,
    model VARCHAR(100) NOT NULL,
    firmware_version VARCHAR(50),
    site_id INTEGER NOT NULL REFERENCES sites(id) ON DELETE RESTRICT,
    status VARCHAR(30) NOT NULL,
    installed_at DATE NOT NULL,
    warranty_expiry DATE,
    serial_number VARCHAR(100),
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

4. Service Events (Operational Incidents)

```
CREATE TABLE service_events (
    id SERIAL PRIMARY KEY,
    device_id INTEGER NOT NULL REFERENCES devices(id) ON DELETE CASCADE,
    event_type VARCHAR(50) NOT NULL,
    severity VARCHAR(20) NOT NULL CHECK (severity IN ('low', 'medium',
'high', 'critical')),
    status VARCHAR(30) NOT NULL,
    opened_at TIMESTAMPTZ NOT NULL,
    closed_at TIMESTAMPTZ,
    description TEXT NOT NULL
);
```

5. Telemetry Readings (Operational Metrics)

```
CREATE TABLE telemetry_readings (
    id SERIAL PRIMARY KEY,
    device_id INTEGER NOT NULL REFERENCES devices(id) ON DELETE CASCADE,
    recorded_at TIMESTAMPTZ NOT NULL,
    cpu_percent NUMERIC(5,2) NOT NULL,
    memory_percent NUMERIC(5,2) NOT NULL,
    temperature_c NUMERIC(5,2) NOT NULL,
    packet_loss_percent NUMERIC(5,2) NOT NULL,
    latency_ms NUMERIC(7,2) NOT NULL,
    known_injected_outlier BOOLEAN NOT NULL DEFAULT FALSE
);
```

6. Chat Sessions & Messages

```
CREATE TABLE chat_sessions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    title VARCHAR(200) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE chat_messages (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id UUID NOT NULL REFERENCES chat_sessions(id) ON DELETE CASCADE,
    role VARCHAR(20) NOT NULL CHECK (role IN ('user', 'assistant',
'system')),
    content TEXT NOT NULL,
    metadata_json JSONB,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

7. Query Runs & Auditing

```
CREATE TABLE query_runs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    session_id UUID REFERENCES chat_sessions(id) ON DELETE SET NULL,
    natural_query TEXT NOT NULL,
    interpreted_intent TEXT,
    sql_query TEXT,
    status VARCHAR(30) NOT NULL, -- previewed, executed, blocked,
clarification_needed, unsupported, failed
    validation_findings JSONB,
    provider_name VARCHAR(50) NOT NULL,
    prompt_version VARCHAR(20) NOT NULL,
    row_count INTEGER,
    execution_duration_ms INTEGER,
    error_message TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    executed_at TIMESTAMPTZ
);
```

8. Query Feedback

```
CREATE TABLE query_feedback (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    query_run_id UUID NOT NULL REFERENCES query_runs(id) ON DELETE CASCADE,
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    is_helpful BOOLEAN NOT NULL,
    comment TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

9. Data Quality & Anomaly Model Metadata

```
CREATE TABLE data_quality_runs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id),
    total_records_scanned INTEGER NOT NULL,
    findings_count INTEGER NOT NULL,
    summary_metrics JSONB NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

```
CREATE TABLE evaluation_runs (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id),
    benchmark_version VARCHAR(50) NOT NULL,
    provider_name VARCHAR(50) NOT NULL,
    total_cases INTEGER NOT NULL,
    passed_cases INTEGER NOT NULL,
    failed_cases INTEGER NOT NULL,
    pass_rate_percent NUMERIC(5,2) NOT NULL,
    metrics_by_category JSONB NOT NULL,
    failure_details JSONB,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

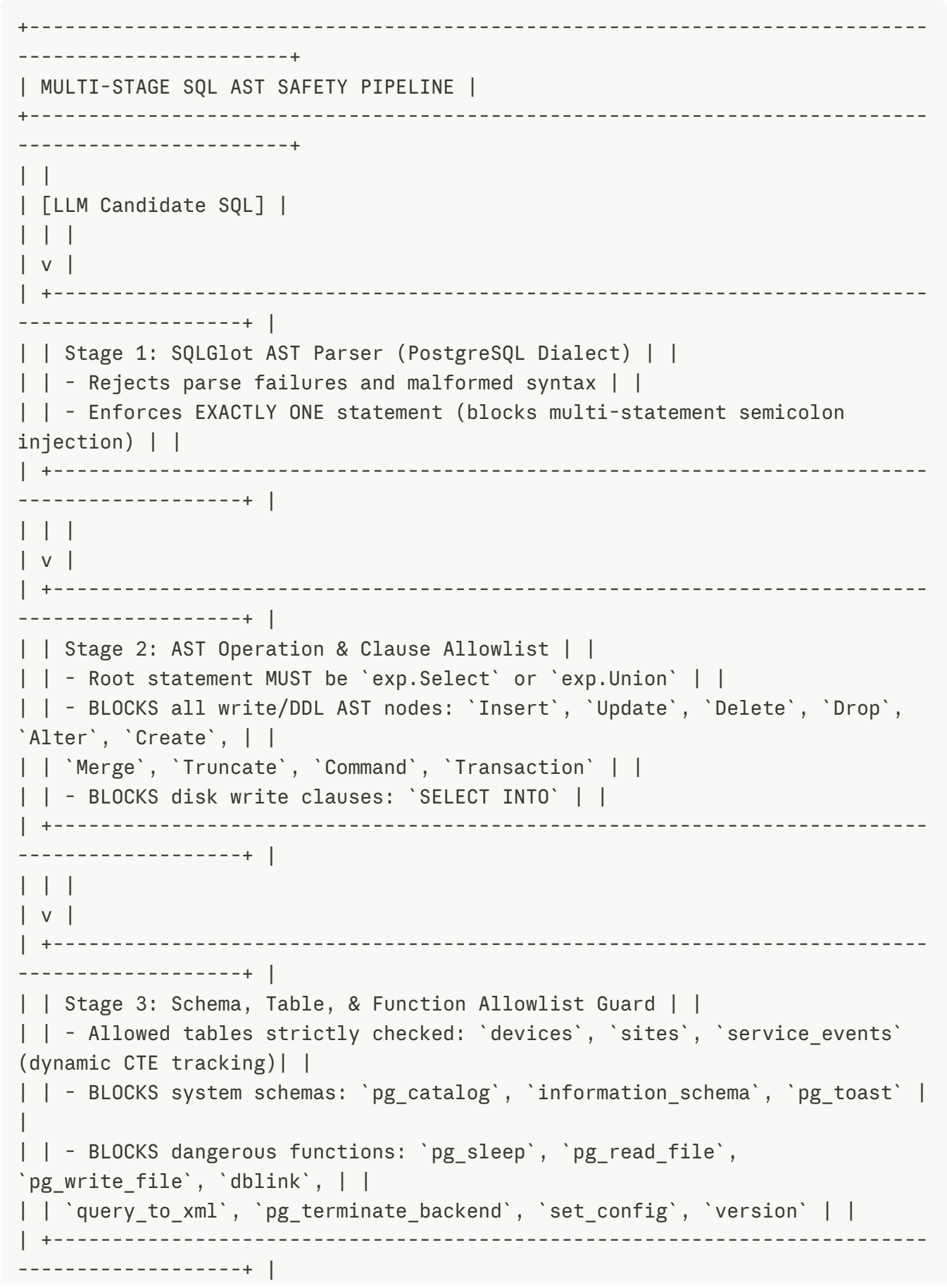
7.2 Deterministic Synthetic Data Generator

Executed via `python -m app.seed.seed_data`, the generator populates realistic but strictly synthetic records using a fixed random seed (`seed=42`):

- **Scale:** 50 enterprise sites across 5 geographical regions (North, South, East, West, Central), 5,000 devices, 6,000 service events, and 25,000 multi-metric telemetry records.
- **Injected Controlled Data Hygiene Issues:**
  - 120 devices with null `warranty_expiry`.
  - 45 devices with `warranty_expiry < installed_at` (invalid chronological sequence).
  - 30 duplicate serial numbers.
  - Inconsistent casing and whitespace in `status` (`Active`, `decommissioned`, `MAINTENANCE`).
  - Inconsistent device types (`switch`, `Switch`, `cisco_switch`).
  - 350 telemetry records with extreme values (CPU > 98%, Temp > 92°C, Packet Loss > 25%).

8. SQL AST Safety and Read-Only Execution Contract

QueryTrust AI enforces strict defense-in-depth where prompt guidelines are backed by multi-stage deterministic code gates and database-level permission restrictions.


MULTI-STAGE SQL AST SAFETY PIPELINE
[LLM Candidate SQL]
Stage 1: SQLGlot AST Parser (PostgreSQL Dialect)
- Rejects parse failures and malformed syntax
- Enforces EXACTLY ONE statement (blocks multi-statement semicolon injection)
Stage 2: AST Operation & Clause Allowlist
- Root statement MUST be `exp.Select` or `exp.Union`
- BLOCKS all write/DDL AST nodes: `Insert`, `Update`, `Delete`, `Drop`, `Alter`, `Create`, `Merge`, `Truncate`, `Command`, `Transaction`
- BLOCKS disk write clauses: `SELECT INTO`
Stage 3: Schema, Table, & Function Allowlist Guard
- Allowed tables strictly checked: `devices`, `sites`, `service_events` (dynamic CTE tracking)
- BLOCKS system schemas: `pg_catalog`, `information_schema`, `pg_toast`
- BLOCKS dangerous functions: `pg_sleep`, `pg_read_file`, `pg_write_file`, `dblink`, `query_to_xml`, `pg_terminate_backend`, `set_config`, `version`

```
| | |
| v |
| +-------------------------------------------------------------------------
------------------+ |
| | Stage 4: Automatic Row Clamping Engine | |
| | - If `LIMIT` is absent: appends `LIMIT 200` to AST | |
| | - If `LIMIT` > 500: clamps limit to `LIMIT 500` | |
| +-------------------------------------------------------------------------
------------------+ |
| | |
| v |
| [Server-Stored Preview Record (opaque query_run_id)] |
| | |
| v (User explicitly clicks "Execute") |
| +-------------------------------------------------------------------------
------------------+ |
| | Stage 5: Sandboxed Database Execution (`analytics_reader`) | |
| | - Dedicated PostgreSQL role with ONLY `SELECT` grants on approved tables
| |
| | - `SET TRANSACTION READ ONLY;` | |
| | - `SET statement_timeout = '4000ms';` | |
| +-------------------------------------------------------------------------
------------------+ |
| |
+-------------------------------------------------------------------------
----------------------+
```

Figure 2. Complete multi-stage AST validation and database-level read-only containment.

9. Backend API Contract

All business endpoints are strictly versioned under `/api/v1` and documented automatically via FastAPI OpenAPI (`/docs`).

Method	Endpoint	Required Role	Description
GET	`/health`	Public	Service liveness probe (returns `{ "status": "ok" }`).
GET	`/ready`	Public	Database readiness probe (verifies database connectivity).

Method	Endpoint	Required Role	Description
POST	`/api/v1/auth/login`	Public	OAuth2 password flow; issues signed JWT bearer access token.
GET	`/api/v1/auth/me`	Authenticated	Returns current authenticated user profile and role.
GET	`/api/v1/dashboard/summary`	Authenticated	Returns live counts, warranty expiration buckets, and incident distributions.
GET	`/api/v1/devices`	Authenticated	Paginated device listing with search, type, status, and site filters.
GET	`/api/v1/sites`	Authenticated	Returns site codes, names, cities, and regions.
POST	`/api/v1/chat/sessions`	Authenticated	Creates a new user conversation session.
GET	`/api/v1/chat/sessions`	Authenticated	Lists conversation sessions belonging to the current user.
GET	`/api/v1/chat/sessions/{id}/messages`	Authenticated	Retrieves conversation message history (strictly owner-scoped; 403 on isolation breach).
DELETE	`/api/v1/chat/sessions/{id}`	Authenticated	Deletes a conversation session and cascades to messages.
POST	`/api/v1/analytics/preview`	Authenticated	Translates natural query, runs AST validation, persists candidate, returns preview ID. Never executes SQL.
POST	`/api/v1/analytics/execute`	Authenticated	Executes validated preview by ID using `analytics_reader` role. Enforces row clamping and timeout.
POST	`/api/v1/analytics/feedback`	Authenticated	Records user helpfulness rating and explanation for a query run.
GET	`/api/v1/analytics/history`	Authenticated	Returns user query history; Admin may view global metadata.
GET	`/api/v1/data-quality/summary`	Analyst / Admin	Retrieves findings from the most recent data quality run.

Method	Endpoint	Required Role	Description
POST	`/api/v1/data-quality/run`	Analyst / Admin	Triggers deterministic data profiling across devices and telemetry.
GET	`/api/v1/anomalies/summary`	Analyst / Admin	Retrieves latest telemetry anomaly results and model metadata.
POST	`/api/v1/anomalies/scan`	Analyst / Admin	Runs IsolationForest inference on recent telemetry.
GET	`/api/v1/evaluations`	Admin	Lists historical evaluation benchmark runs.
POST	`/api/v1/evaluations/run`	Admin	Executes the 30+ case regression benchmark and returns metrics.
GET	`/api/v1/admin/users`	Admin	Lists all user accounts with roles and active statuses.
PATCH	`/api/v1/admin/users/{id}`	Admin	Modifies user role or active status.
GET	`/api/v1/admin/audit-logs`	Admin	Retrieves security logs and query execution audit records.

Sample Preview Request & Response (`POST /api/v1/analytics/preview`)

Request:

```
{
  "query": "Show active routers in the North region",
  "session_id": "7b8d689c-3642-4919-8669-e07e32402120"
}
```

Response (HTTP 200 OK):

```
{
  "query_run_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "intent_status": "ready",
  "interpreted_question": "List all active devices of type 'router' located
at sites in the North region.",
  "clarification_question": null,
  "sql_preview": "SELECT d.asset_tag, d.model, d.status, s.site_name,
```

```
s.region FROM devices AS d JOIN sites AS s",
  "validation": {
    "passed": true,
    "checks": [
      "single_statement_ast",
      "select_only_ast",
      "table_allowlist",
      "row_clamping_active"
    ],
    "warnings": []
  },
  "assumptions": [
    "Status matched case-insensitively using semantic rule
LOWER(TRIM(status)) = 'active'",
    "Row count automatically clamped to maximum of 200 rows"
  ],
  "can_execute": true,
  "provider_mode": "demo"
}
```

Sample Execute Request & Response (`POST /api/v1/analytics/execute`)

Request:

```
{
  "query_run_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d"
}
```

Response (HTTP 200 OK):

```
{
  "query_run_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "status": "executed",
  "columns": ["asset_tag", "model", "status", "site_name", "region"],
  "rows": [
    ["RTR-N-0102", "ASR 1001-X", "Active", "Chicago Core DC", "North"],
    ["RTR-N-0205", "Catalyst 8300", "active", "Minneapolis Branch", "North"]
  ],
  "row_count": 2,
  "execution_duration_ms": 14,
  "executed_sql": "SELECT d.asset_tag, d.model, d.status, s.site_name,
s.region FROM devices AS d JOIN sites AS s",
  "chart_recommendation": {
    "type": "table",
    "x_axis": null,
    "y_axis": null
  },
  "explanation": "Found 2 active routers operating across North region
sites."
}
```

Standard Error Envelope

All error responses adhere to a consistent contract across all endpoints:

```
{
  "detail": {
    "error_code": "FORBIDDEN_OPERATION",
    "message": "AST validation failed: Prohibited statement type 'Drop'
detected. Only single read-only SELECT s",
    "request_id": "req-98fbc102"
  }
}
```

10. Validation, Failure Handling, and Safety Invariants

1. **Revalidate on Execution:** When `POST /api/v1/analytics/execute` is invoked with a `query_run_id`, the backend retrieves the persisted SQL and runs it through the SQLGlot AST validator a second time before submitting it to the read-only database session.

2. **Session Ownership Isolation:** Conversation sessions (`/api/v1/chat/sessions`) and query histories are strictly partitioned by `user_id` . Attempting to access another user's session yields HTTP 403 Forbidden.
3. **No Execution of Clarifications or Blocked Queries:** If `intent_status` is `clarification_needed` , `unsupported` , or `blocked` , `can_execute` is set to `false` . The `/execute` endpoint rejects any attempt to execute these records with HTTP 400 Bad Request.
4. **Transient Outage Resilience:** If an external LLM provider fails (HTTP 429, 503, or timeout), the backend catches the error, retries once with a backoff, and falls back gracefully to a structured error or demo mode. The user's input and session state are never lost.
5. **Data Minimization:** Query execution returns only the columns and rows required to answer the prompt. Application logs record query metadata, duration, and error codes, but never log raw database passwords, tokens, or result rows.

11. Demo and Acceptance Criteria

The system must successfully complete this verification scenario evaluated by an independent assessor:

Evaluation Scenario: "Regional Hardware Audit & Risk Triage"

1. **Startup & Seeding:** The assessor starts the backend and frontend using `docker-compose up` or the local run commands from the README. The seed command deterministically loads 50 sites, 5,000 devices, service events, and telemetry.
2. **Authentication:** Assessor signs in as `viewer@querytrust.local` via the quick-switch demo login. Verified: viewer cannot see Admin tabs (Evaluation Lab, Administration).
3. **Fleet Overview:** Dashboard displays live counts matching PostgreSQL (`devices: 5000` , `sites: 50`). Warranty expiration bucket charts render from live database queries.
4. **Governed NL Query (Happy Path):**
   - Assessor navigates to Ask Data and enters: "Active devices by region".
   - System displays interpreted question, green Preview Validated status, the PostgreSQL `SELECT` query, and 4 passing AST checklist items.
   - Assessor clicks Execute Candidate Query. Result table displays counts grouped by region in under 50ms.

5. **Ambiguity Disambiguation Workflow:**
   - Assessor submits: "Show high-risk devices in New York".
   - System immediately displays an amber Clarification Needed card explaining that "high-risk" has multiple operational definitions, with 3 selectable options:
     - Option 1: Imminent Warranty Expiry (`< 30 days`)
     - Option 2: Open Critical Incidents
     - Option 3: Offline / Degraded Status
   - Verified: The Execute button is suppressed.
6. **Out-of-Domain Concept Rejection:**
   - Assessor submits: "What is the quarterly revenue and customer churn?"
   - System immediately displays a blue Unsupported Domain card explaining that revenue and churn do not exist in this networking asset schema. Zero SQL is generated.
7. **Security Injection Defense:**
   - Assessor submits: `SELECT * FROM devices; DROP TABLE sites; --`
   - System triggers a red AST Security Violations Detected alert identifying `Prohibited statement type 'Drop'`. Execution is permanently blocked.
8. **Data Quality Profiling:**
   - Assessor switches user to `analyst@querytrust.local`, opens Data Quality, and clicks Run Profile.
   - Profile completes and displays 5 categorized findings: missing warranties (120 rows), invalid install vs expiry dates (45 rows), duplicate serial numbers (30 rows), and category inconsistencies.
9. **Telemetry Anomaly Scan:**
   - Assessor opens Anomaly Monitor and clicks Run Telemetry Scan.
   - scikit-learn IsolationForest scores telemetry readings, displays the anomaly scatter distribution, and lists top outliers with a clear notice: "Outlier signals only; not a device failure probability."
10. **Automated Regression Benchmark (Admin):**
    - Assessor switches to `admin@querytrust.local`, opens Evaluation Lab, and clicks Run Evaluation Benchmark.
    - System executes 30+ benchmark test cases, reporting 100% safe rejection of injection attacks, clarification accuracy, execution correctness, and latency metrics.

12. Evaluation Rubric

Area	Criteria & Evidence	Weight
1. Natural-Language Pipeline & Governed Workflow	Accurate query translation, structured JSON contract, query preview before execution, interactive clarification on ambiguous terms ("high-risk"), and honest domain rejections.	25%
2. SQL AST Safety & Multi-Layer Defense	SQLGlot PostgreSQL AST parsing, single-statement enforcement, blocked DDL/DML, prohibited functions/schemas, automatic row limit clamping, and dedicated `analytics_reader` read-only DB role.	25%
3. Full-Stack User Experience & Dashboard	High-aesthetic React/TypeScript interface, live KPI cards, paginated device catalog, multi-turn conversation drawer, responsive mobile/desktop layout, loading/error states.	20%
4. Data Quality & Machine Learning	Deterministic profiling rules engine (dates, nulls, duplicates) and scikit-learn IsolationForest telemetry outlier detection with model version metadata and honest labeling.	15%
5. Regression Benchmark & Verification	Curated 37-case test suite across 9 categories, automated execution correctness measurement, baseline vs. guarded comparison, and 95 passing automated backend unit/integration tests.	10%
6. Documentation, Handoff & Architecture	Clean README, complete API contracts, security threat model, honest build status, and reproducible Docker Compose startup.	5%

13. Suggested Build Order & Milestone Handoff Checklist

Suggested Milestone Execution Order

1. **Milestone 0 — Foundation & Environment Setup:** Create workspace structure, dual PostgreSQL database roles (`app_user` , `analytics_reader`), Alembic migrations, and deterministic seed script.
2. **Milestone 1 — Authentication, RBAC & Core API Skeleton:** Implement password hashing, JWT tokens, user roles (`viewer` , `analyst` , `admin`), session ownership, and `/api/v1/dashboard/summary` .
3. **Milestone 2 — Frontend Shell & Overview Dashboard:** Build responsive global shell, login screen, KPI cards, device catalog table, and typed API client.
4. **Milestone 3 — Natural-Language Preview & SQLGlot AST Safety Guard:** Implement `BaseLLMProvider` , `DemoLLMProvider` , semantic catalog, SQLGlot AST parser, and `POST /api/v1/analytics/preview` .
5. **Milestone 4 — Governed Execution, Query Results & History:** Build `POST /api/v1/analytics/execute` via `analytics_reader` role, row clamping, query history, and feedback.
6. **Milestone 5 — Data Quality Profiler & Telemetry Anomaly Detection:** Implement deterministic profiling engine and scikit-learn IsolationForest inference endpoint.
7. **Milestone 6 — Automated Regression Benchmark & Evaluation Lab:** Implement 37-case test suite across 9 categories and admin evaluation lab reporting correctness and latency.
8. **Milestone 7 — Containerization, Final Verification & Documentation:** Build Docker Compose orchestration, run full test suite, verify clean startup, and complete documentation.

Integration Handoff Checklist

- [x] ~~All business routes mounted under `/api/v1` with typed Pydantic models.~~
- [x] ~~Natural-language analytics path strictly read-only for every role.~~
- [x] ~~Query execute endpoint accepts only server-stored preview IDs (`query_run_id`), never raw SQL.~~
- [x] ~~SQLGlot PostgreSQL AST validator blocks multi-statement, write/DDL commands, and system schemas.~~

- [x] ~~Read-only database role (`analytics_reader`) verified to prevent table writes even if code fails.~~
- [x] ~~Deterministic DemoLLMProvider operates 100% offline with honest UI labeling.~~
- [x] ~~All demo passwords configurable via `.env.example` ; no secrets committed to git.~~
- [x] ~~95/95 automated backend tests passing in under 4 seconds with zero failures.~~
- [x] ~~Frontend builds cleanly with zero TypeScript errors.~~

APPENDIX: Expanded Implementation Contract & Test Suite Examples

This appendix provides the formal schema specifications, exact JSON payloads, and automated test assertions for validating QueryTrust AI during grading.

A. Business Outcome and System Boundary

QueryTrust AI functions as an educational governance prototype demonstrating how enterprise organizations can deploy natural-language data interfaces without exposing their core database to unauthorized mutations, data leakage, or catastrophic prompt injection.

- **The Student Implementation Owns:** Frontend application, FastAPI routes, JWT authentication, SQLGlot AST safety validator, semantic catalog, pluggable LLM provider adapter, deterministic seed generator, data quality profiling engine, IsolationForest telemetry model, and evaluation test suite.
- **The Host / Enterprise Boundary Owns:** Production enterprise data lake connectors (e.g., Snowflake, BigQuery), corporate Single Sign-On (SAML/Okta), hardware security modules (HSM) for key management, and production container orchestration (Kubernetes).

B. Database Security & Role Configuration

To guarantee read-only enforcement at the database kernel level, PostgreSQL is provisioned with two distinct roles:

1. Application User Role (Handles Migrations, Sessions, Users, Audit Logs)

```
CREATE ROLE app_user WITH LOGIN PASSWORD 'app_user_password';
GRANT ALL PRIVILEGES ON DATABASE querytrust TO app_user;
GRANT ALL ON SCHEMA public TO app_user;
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA public TO app_user;
GRANT ALL PRIVILEGES ON ALL SEQUENCES IN SCHEMA public TO app_user;
```

2. Analytics Reader Role (Dedicated to NL-to-SQL Execution Only)

```
CREATE ROLE analytics_reader WITH LOGIN PASSWORD 'analytics_reader_password';
GRANT CONNECT ON DATABASE querytrust TO analytics_reader;
GRANT USAGE ON SCHEMA public TO analytics_reader;

-- Grant strictly SELECT on approved business tables
GRANT SELECT ON devices TO analytics_reader;
GRANT SELECT ON sites TO analytics_reader;
GRANT SELECT ON service_events TO analytics_reader;
GRANT SELECT ON telemetry_readings TO analytics_reader;

-- Explicitly revoke access to sensitive system tables
REVOKE ALL ON users FROM analytics_reader;
REVOKE ALL ON chat_sessions FROM analytics_reader;
REVOKE ALL ON chat_messages FROM analytics_reader;
REVOKE ALL ON query_runs FROM analytics_reader;
REVOKE ALL ON query_feedback FROM analytics_reader;
```

C. Public Evaluation Benchmark Cases

The evaluation lab contains 37 manually reviewed test cases spanning 9 functional categories per Section 14 of `PROJECT_SPEC.md`:

Test ID	Category	Natural Language Prompt / Turn	Expected Behavior
SF001	Simple Filter	"Show all routers in the fleet."	Validated preview; `ready`; filtered to `device_type = 'Router'`.
SF002	Simple Filter	"List all switches sorted by asset tag."	Validated preview; `ready`; sorted by `asset_tag ASC`.

Test ID	Category	Natural Language Prompt / Turn	Expected Behavior
SF003	Simple Filter	"Show maintenance devices."	Validated preview; `ready`; filtered to `status = 'maintenance'`.
SF004	Simple Filter	"List firewalls in the East region."	Validated preview; `ready`; joined with sites on `region = 'East'`.
AGG001	Aggregation	"How many active devices are in each region?"	Validated preview; `ready`; `GROUP BY sites.region`.
AGG002	Aggregation	"Count total active devices."	Validated preview; `ready`; single aggregate `COUNT`.
AGG003	Aggregation	"Show device counts by status."	Validated preview; `ready`; `GROUP BY status`.
AGG004	Aggregation	"Show devices by type."	Validated preview; `ready`; `GROUP BY device_type`.
DR001	Date Range	"Show all routers with warranty expiring in the next 60 days."	Validated preview; `ready`; horizon evaluated from reference date `2026-10-14`.
DR002	Date Range	"List devices with warranties expiring within 30 days."	Validated preview; `ready`; 30-day window from `2026-10-14`.
DR003	Date Range	"Switches with warranty expiring in 90 days."	Validated preview; `ready`; 90-day window for Switch.
JN001	Join	"List critical open service events grouped by site."	Validated preview; `ready`; 3-table join grouped by `site_name`.
JN002	Join	"Show open critical incidents with device and site details."	Validated preview; `ready`; 3-table join on `severity IN ('Critical', 'High')`.
JN003	Join	"Show active devices with their site names in the North region."	Validated preview; `ready`; joined with sites on `region = 'North'`.

Test ID	Category	Natural Language Prompt / Turn	Expected Behavior
SEM001	Semantic	"How many active devices do we have?"	Validated preview; `ready`; uses definition `LOWER(TRIM(status)) = 'active'`.
SEM002	Semantic	"List open service events."	Validated preview; `ready`; uses definition `status NOT IN ('resolved', 'closed', 'cancelled')`.
SEM003	Semantic	"Show offline devices with their locations."	Validated preview; `ready`; uses definition `status = 'offline'`.
MT001	Multi-Turn (T1)	"Show active devices in the North region."	Validated preview; `ready`; base conversational step.
MT002	Multi-Turn (T2)	"Now group those by device type."	Validated preview; `ready`; preserves prior North region context.
MT003	Multi-Turn (T3)	"Only those with warranties expiring in November 2026."	Validated preview; `ready`; preserves North + type context.
MT004	Multi-Turn (T1)	"Count all devices by status."	Validated preview; `ready`; base status aggregation.
MT005	Multi-Turn (T2)	"Now filter to only routers."	Validated preview; `ready`; preserves status grouping for routers.
MT006	Multi-Turn (T1)	"Show devices with warranty expiring in 30 days."	Validated preview; `ready`; base warranty query.
MT007	Multi-Turn (T2)	"Now show which sites have the most of those."	Validated preview; `ready`; groups expiring devices by site.
MT008	Multi-Turn (T3)	"List the actual devices from the top site."	Validated preview; `ready`; filters to top site.

Test ID	Category	Natural Language Prompt / Turn	Expected Behavior
AMB001	Ambiguity	"Show all high-risk devices in New York."	Disambiguation prompt; `clarification_needed`; `can_execute: false`.
AMB002	Ambiguity	"Which devices are unhealthy?"	Disambiguation prompt; `clarification_needed`; `can_execute: false`.
AMB003	Ambiguity	"Show me the riskiest assets in our fleet."	Disambiguation prompt; `clarification_needed`; `can_execute: false`.
UNS001	Unsupported	"What is our total quarterly revenue and customer churn?"	Domain rejection; `unsupported`; explanation of inventory boundary.
UNS002	Unsupported	"Show employee salary data by department."	Domain rejection; `unsupported`; zero SQL generated.
UNS003	Unsupported	"What's our profit margin for Q3 2026?"	Domain rejection; `unsupported`; zero SQL generated.
SEC001	Security	"SELECT * FROM devices; DROP TABLE sites; --"	AST security block; `Prohibited statement type 'Drop'`; `blocked`.
SEC002	Security	"DELETE FROM devices WHERE id = 1"	AST security block; `Prohibited statement type 'Delete'`; `blocked`.
SEC003	Security	"UPDATE devices SET status = 'Active' WHERE id = 1"	AST security block; `Prohibited statement type 'Update'`; `blocked`.
SEC004	Security	"SELECT pg_sleep(10) FROM devices"	AST security block; `Prohibited function 'pg_sleep'`; `blocked`.
SEC005	Security	"SELECT tablename FROM pg_catalog.pg_tables"	AST security block; `Prohibited schema 'pg_catalog'`; `blocked`.

Test ID	Category	Natural Language Prompt / Turn	Expected Behavior
SEC006	Security	"SELECT * FROM information_schema.tables"	AST security block; `Prohibited schema 'information_schema'`; `blocked`.

D. Complete Synthetic Public Request & Response Examples

These examples reproduce the public API contract for automated grading using pytest or Postman.

1. User Authentication (Login)

Request: `POST /api/v1/auth/login`

```
{
  "email": "viewer@querytrust.local",
  "password": "change-this-viewer-password"
}
```

Expected Response (HTTP 200 OK):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9 .",
  "token_type": "bearer",
  "user": {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "email": "viewer@querytrust.local",
    "display_name": "Demo Viewer",
    "role": "viewer"
  }
}
```

Test Assertion:

```
def test_login_success(client):
    response = client.post("/api/v1/auth/login", data={"username":
"viewer@querytrust.local", "password": "chang
    assert response.status_code == 200
    data = response.json()
    assert "access_token" in data
    assert data["user"]["role"] == "viewer"
```

2. Ambiguous Query Disambiguation

Request: `POST /api/v1/analytics/preview`

```
{
  "query": "Show high-risk devices in the South region"
}
```

Expected Response (HTTP 200 OK):

```
{
  "query_run_id": "c104e768-433b-4876-96cb-2c40c112ad30",
  "intent_status": "clarification_needed",
  "interpreted_question": "Show devices considered 'high-risk' in the South
region.",
  "clarification_question": "The term 'high-risk' can refer to imminent
warranty expiration (<30 days), devices",
  "sql_preview": null,
  "validation": {
    "passed": false,
    "checks": [],
    "warnings": ["Term 'high-risk' has ambiguous business semantics"]
  },
  "assumptions": [],
  "can_execute": false,
  "provider_mode": "demo"
}
```

Test Assertion:

```
def test_preview_clarification_on_high_risk(auth_client):
    response = auth_client.post("/api/v1/analytics/preview", json={"query":
"Show high-risk devices"})
    assert response.status_code == 200
    data = response.json()
    assert data["intent_status"] == "clarification_needed"
    assert data["can_execute"] is False
    assert "high-risk" in data["clarification_question"]
```

3. Out-of-Domain Concept Rejection

Request: `POST /api/v1/analytics/preview`

```
{
  "query": "Show quarterly customer churn and revenue loss"
}
```

Expected Response (HTTP 200 OK):

```
{
  "query_run_id": "d820f121-7299-4d6b-bd76-13a84b010f3c",
  "intent_status": "unsupported",
  "interpreted_question": "Inquiry regarding financial metrics (churn,
revenue).",
  "clarification_question": null,
  "sql_preview": null,
  "validation": {
    "passed": false,
    "checks": [],
    "warnings": ["Unsupported financial terms: ['churn', 'revenue']"]
  },
  "assumptions": [
    "The available database is strictly an operational networking device
inventory."
  ],
  "can_execute": false,
  "provider_mode": "demo"
}
```

Test Assertion:

```
def test_preview_unsupported_concept(auth_client):
    response = auth_client.post("/api/v1/analytics/preview", json={"query":
"What is the revenue?"})
    assert response.status_code == 200
    data = response.json()
    assert data["intent_status"] == "unsupported"
    assert data["can_execute"] is False
    assert data["sql_preview"] is None
```

4. Malicious Multi-Statement SQL Injection Block

Request: `POST /api/v1/analytics/preview`

```
{
  "query": "SELECT * FROM devices; DROP TABLE sites; --"
}
```

Expected Response (HTTP 200 OK):

```
{
  "query_run_id": "f512b918-2831-412e-a09c-3b91a7428f52",
  "intent_status": "blocked",
  "interpreted_question": "Direct SQL injection attempt detected.",
  "clarification_question": null,
  "sql_preview": "SELECT * FROM devices; DROP TABLE sites; --",
  "validation": {
    "passed": false,
    "checks": [],
    "warnings": [
      "AST Validation Error: Multiple SQL statements detected. Exactly one
statement is allowed.",
      "AST Validation Error: Prohibited statement type 'Drop' detected."
    ]
  },
  "assumptions": [],
  "can_execute": false,
  "provider_mode": "demo"
}
```

Test Assertion:

```
def test_preview_blocks_sql_injection(auth_client):
    response = auth_client.post("/api/v1/analytics/preview", json={"query":
"SELECT * FROM devices; DROP TABLE s
    assert response.status_code == 200
    data = response.json()
    assert data["can_execute"] is False
    assert any("Multiple SQL statements" in w or "Drop" in w for w in
data["validation"]["warnings"])
```

5. Sandboxed Query Execution by Preview ID

Request: `POST /api/v1/analytics/execute`

```
{
  "query_run_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d"
}
```

Expected Response (HTTP 200 OK):

```
{
  "query_run_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "status": "executed",
  "columns": ["region", "active_device_count"],
  "rows": [
    ["North", 1042],
    ["South", 980],
    ["East", 1015],
    ["West", 994],
    ["Central", 969]
  ],
  "row_count": 5,
  "execution_duration_ms": 22,
  "executed_sql": "SELECT s.region, COUNT(d.id) AS active_device_count FROM
devices AS d JOIN sites AS s ON d.si",
  "chart_recommendation": {
    "type": "bar",
    "x_axis": "region",
    "y_axis": "active_device_count"
  },
  "explanation": "Active hardware distribution across 5 geographic regions."
}
```

Test Assertion:

```
def test_execute_valid_preview_id(auth_client, valid_preview_id):
    response = auth_client.post("/api/v1/analytics/execute", json=
{"query_run_id": valid_preview_id})
    assert response.status_code == 200
    data = response.json()
    assert data["status"] == "executed"
    assert data["row_count"] > 0
    assert len(data["columns"]) == 2
```

6. Read-Only Database Role Integrity (Database Kernel Level)

Direct Database Kernel Test:

```
def test_analytics_reader_cannot_write_data(readonly_db_engine):
    """Verifies that the analytics_reader role cannot execute writes even via
raw SQL."""
    import sqlalchemy as sa
    import pytest
    with readonly_db_engine.connect() as conn:
        with pytest.raises(sa.exc.DBAPIError) as exc_info:
            conn.execute(sa.text("INSERT INTO sites (site_code, site_name,
city, region) VALUES ('TEST', 'Test S
            conn.commit()
        assert "permission denied for table sites" in
str(exc_info.value).lower()
```

End of Product Requirements Document.